

Date	23 July 2026
Project Name	SB-STOCKS
Maximum Marks	5

Project Documentation

Project Overview

SB-STOCKS is a full-stack MERN application that allows users to practice stock trading using virtual funds in a simulated market environment.

System Architecture

React frontend connects to an Express API running on Node.js. Data is stored securely in MongoDB.

Database Schema

- Users: { name, email, password, role, virtualBalance, createdAt }
- Stocks: { symbol, name, currentPrice, previousClose, change, volume, sector }
- Portfolios: { userId, stockId, quantity, averageBuyPrice, currentValue }
- Transactions: { userId, stockId, type(buy/sell), quantity, price, totalAmount, date }
- Watchlists: { userId, stockId, addedAt }

API Endpoints

Method	Endpoint	Description
POST	/api/auth/register	Register new user
POST	/api/auth/login	Login user
GET	/api/stocks	Get all stocks
GET	/api/stocks/:id	Get stock by ID
POST	/api/stocks	Add stock (Admin)
PUT	/api/stocks/:id	Update stock (Admin)
DELETE	/api/stocks/:id	Delete stock (Admin)
POST	/api/trade/buy	Buy stocks
POST	/api/trade/sell	Sell stocks
GET	/api/portfolio	Get user portfolio
GET	/api/transactions	Get transaction history



GET	/api/watchlist	Get user watchlist
POST	/api/watchlist	Add to watchlist
DELETE	/api/watchlist/:id	Remove from watchlist



Setup Instructions & Screenshots

Setup follows standard MERN application procedures. Run npm install in both client and server directories.

(Screenshots placeholder space)

Conclusion:

SB-STOCKS successfully delivers a robust, secure, and user-friendly platform for practicing stock trading strategies.